

Cover Page

Data Analysis and Reporting Using Power BI

University Academic Performance, Enrollment, Attendance, Financial, and Department Analytics

Student Information: [Name / ID placeholders]

Course Information: [Course / section placeholders]

Executive Summary

This project creates a complete synthetic university analytics package for Power BI. It includes raw and cleaned datasets, Python generation and validation scripts, data dictionary, relationship documentation, DAX and Power Query libraries, dashboard specifications, report, presentation, and GitHub publication. The cleaned dataset contains 1,000 students, 10 departments, 70 courses, 50 faculty members, 12,500 enrollment records, 22,577 attendance records, and 5,994 payment records.

Introduction

Universities need integrated reporting across enrollment, performance, attendance, payments, faculty workload, and budget utilization. This project demonstrates how related academic tables can be modeled and transformed for a management dashboard in Power BI.

Problem Statement

University management often receives fragmented data from student, academic, finance, and departmental systems. The project addresses this by designing a relational dataset and Power BI-ready semantic model for multi-perspective analytics.

Project Objectives

The objectives are to import and transform multi-table data, define one-to-one, one-to-many, and bridge-resolved many-to-many relationships, create analytical measures, specify meaningful dashboards, and document findings and limitations.

Dataset Description

The dataset covers academic years 2023 to 2026 and is entirely fictional. It contains raw data with controlled quality problems and cleaned data suitable for Power BI modeling. Raw and cleaned workbooks are stored separately and are also exported as individual CSV files.

Table	Rows	Columns
Students	1,000	12
StudentProfile	1,000	7
Departments	10	6
Courses	70	7
Faculty	50	7
Semesters	11	5
Enrollment	12,500	14
Attendance	22,577	6
Payments	5,994	10
DepartmentBudget	40	11

Table Descriptions

Students stores student demographic and admission attributes. StudentProfile stores one-to-one profile details. Departments, Courses, Faculty, and Semesters are dimensions. Enrollment is both transaction fact and bridge table. Attendance, Payments, and DepartmentBudget are fact tables.

Data Dictionary

The complete data dictionary is available in documentation/University_Data_Dictionary.xlsx. It records table names, column names, data types, purpose, and null counts.

Data Import Process

Open Power BI Desktop, select Get Data, choose Excel Workbook, select data/cleaned/University_Academic_Analytics_Cleaned.xlsx, load all worksheets, open Transform Data, review headers, confirm types, apply transformations, load data, open Model View, define relationships, hide technical identifiers where appropriate, create _Measures, create the Date table, and mark it as the official Date table.

Data-Cleaning Process

The cleaning process standardizes text, removes duplicates, corrects data types, validates marks and attendance percentages, recalculates payment balances, fills controlled null values where appropriate, and creates academic, attendance, payment, and budget risk categories.

Power Query Transformations

Detailed Power Query M code is provided in powerbi/Power_Query_M_Code.md. It includes promoting headers, renaming/typing columns, trimming and cleaning text, replacing inconsistent values, handling nulls, validating marks, converting text dates, calculating outstanding payments, and creating conditional categories.

Data-Model Design

The model follows a controlled snowflake/star schema. Students, StudentProfile, Departments, Courses, Faculty, and Semesters are dimension-style tables. Enrollment, Attendance, Payments, and DepartmentBudget provide transactional and analytical facts.

One-to-One Relationship

Students[StudentID] relates to StudentProfile[StudentID]. Each StudentID appears once in both tables, allowing profile attributes to extend the student dimension without duplication.

One-to-Many Relationships

Departments filters Students, Courses, Faculty, and DepartmentBudget. Students, Courses, Semesters, and Faculty filter Enrollment. Enrollment filters Attendance. Students and Semesters filter Payments. Relationships are documented in documentation/Relationship_Documentation.md.

Many-to-Many Relationship

Students and Courses are conceptually many-to-many. This is not modeled as an uncontrolled direct relationship. It is resolved through Enrollment: Students -> Enrollment <- Courses.

Bridge-Table Explanation

Enrollment stores the student-course-semester-faculty grain. It resolves many-to-many course participation while also carrying marks, grades, attendance percentage, and pass/fail results.

Star-Schema Explanation

The design keeps descriptive attributes in dimension tables and analytical values in fact/bridge tables. Single-direction filtering is recommended to reduce ambiguity and improve report predictability.

Date Table

The Date table DAX is documented in powerbi/Date_Table_DAX.md. It includes Date, Year, Quarter, QuarterName, MonthNumber, MonthName, ShortMonthName, YearMonth, YearMonthSort, WeekNumber, DayNumber, DayName, AcademicYear, and SemesterPeriod.

DAX Calculated Columns

Calculated column examples are documented in powerbi/DAX_Calculated_Columns.md, including Student Age, Student Age Group, Course Result Category, Graduation Eligibility, and Faculty Experience Group. The report explains when Power Query columns, DAX columns, and measures should be used.

DAX Measures

The project includes 40 DAX measures in powerbi/DAX_Measures.md. They cover student counts, department/course/faculty counts, enrollment metrics, academic performance, pass/fail rates, attendance risk, tuition collection, outstanding amounts, faculty load, growth rates, budget utilization, scholarship rate, ranking, TOPN, dynamic titles, and USERRELATIONSHIP examples.

Dashboard Design

The dashboard is specified as a clean academic management report with consistent typography, limited palette, aligned visuals, accessible contrast, page headings, navigation, slicers, conditional formatting, dynamic titles, and report-page tooltips.

Explanation of Report Pages

The required report pages are Executive Overview, Student Analytics, Academic Performance, Attendance Analytics, Financial Analytics, Department and Faculty Analytics, and an optional Student Drill-Through Detail page. Detailed page specifications are in `documentation/Dashboard_Page_Specifications.md`.

Main Findings

The analysis identifies enrollment concentration by department, courses with higher failure rates, association between attendance and marks, outstanding payment exposure, budget-utilization variation, scholarship distribution, faculty workload monitoring needs, graduation eligibility signals, payment-risk patterns, and semester trend use cases.

Highest enrollment department: English has the largest student count. Recommendation: Review section sizes and faculty allocation. Limitation: Synthetic dataset only.

Highest failure course: C018 has the highest failure rate. Recommendation: Add tutoring and review course design. Limitation: Association only, not causation.

Attendance relationship: Lower attendance bands show lower average final marks. Recommendation: Trigger advising below 75% attendance. Limitation: Cannot prove attendance causes marks.

Outstanding payments: Outstanding balance totals 125,733,250. Recommendation: Segment outreach by payment risk level. Limitation: Synthetic tuition assumptions.

Budget utilization: D007 has the highest utilization in 2023. Recommendation: Review mid-year budget reallocations. Limitation: Budget is generated, not actual finance data.

Scholarship distribution: 100.0% of students receive a scholarship. Recommendation: Track outcomes by scholarship type. Limitation: No real socioeconomic data.

Faculty workload: Enrollment links faculty to taught courses for workload analysis. Recommendation: Monitor distinct courses and enrollments per faculty. Limitation: Workload excludes non-teaching duties.

Graduation eligibility: Credit and GPA measures identify students approaching eligibility. Recommendation: Add drill-through student review page. Limitation: Eligibility rule is simplified.

Payment status: Paid, partial, and unpaid categories provide finance risk visibility. Recommendation: Use respectful outreach workflows. Limitation: Synthetic data includes fictional profiles.

Semester trend: Semester fields support enrollment and performance trend pages. Recommendation: Monitor semester-over-semester growth. Limitation: Trend is generated from random distributions.

Recommendations

Recommended actions include monitoring high-demand departments, adding academic support for high-failure courses, using attendance below 75% as an advising trigger, segmenting payment outreach by risk level, reviewing faculty workload, and tracking budget utilization by department and year.

Limitations

Power BI Desktop was not installed in this environment. Therefore, no .pbix file, actual dashboard screenshots, Power BI Service URL, Power BI App URL, or public embed URL is claimed. The dataset is synthetic, so findings are demonstrations rather than real institutional conclusions.

Future Improvements

Future work should build the .pbix in Power BI Desktop, publish to Power BI Service where licensing permits, add actual screenshots, configure refresh, enable GitHub Pages if desired, and refine report visuals using user feedback.

Conclusion

The project satisfies the data generation, cleaning, validation, modeling documentation, DAX, Power Query, report, presentation, and GitHub repository requirements that can be completed without Power BI Desktop. Remaining work is clearly documented and should be completed in a genuine Power BI authoring environment.

References

Microsoft Power BI documentation, Microsoft DAX documentation, Microsoft Power Query M documentation, and project-generated synthetic data documentation.

Appendix

Appendix files include powerbi/DAX_Measures.md, powerbi/Power_Query_M_Code.md, powerbi/Date_Table_DAX.md, powerbi/DAX_Calculated_Columns.md, documentation/Relationship_Documentation.md, and documentation/Prompt_Compliance_Cross_Check.md.

GitHub Repository Information

Repository: <https://github.com/Monokayser/University-PowerBI-Analytics>

Release: <https://github.com/Monokayser/University-PowerBI-Analytics/releases/tag/v1.0.0>

Power BI Live-Report Information

Power BI Service publication was not completed because Power BI Desktop and a verified authorized Power BI Service session were unavailable. A real URL should be added only after the report is built, published, opened, and tested.